



Assignment 2

Assignment must be completed in groups of 2 members and submitted via Google Classroom by 14.09.2026 at 23:59. If the assignment is not prepared using L^AT_EX, a clear scanned copy of the handwritten work must be uploaded. Ensure that the names and enrollment numbers of all group members are clearly written on the submission. Late submissions will not be accepted.

Question 1: [FDM in Polar Coordinates, 6 Points]

Consider the Laplace equation

$$-\Delta u = f \quad \text{in } \Omega, \quad (1)$$

$$u = g \quad \text{on } \Gamma = \partial\Omega, \quad (2)$$

where $\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1\}$ is the unit disk, and f, g are given smooth functions.

- a) Using polar coordinates $x = r \cos \theta$, $y = r \sin \theta$, show that (1) can be written as

$$-\left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r}\right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}\right) = f(r, \theta), \quad (r, \theta) \in (0, 1) \times (0, 2\pi),$$

with boundary condition $u(1, \theta) = g(\theta)$, $\theta \in [0, 2\pi)$, where g is 2π -periodic in θ .

- b) Construct a second-order finite-difference scheme for the operator

$$\mathcal{L}u := \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r}\right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

on a uniform polar mesh.

Question 2: [Neumann Boundary Conditions, 6 Points]

Consider the Laplace equation

$$-\Delta u = f \quad \text{in } \Omega = (0, 1)^2, \quad (3)$$

$$\partial_{\mathbf{n}} u = g \quad \text{on } \Gamma = \partial\Omega, \quad (4)$$

Derive the finite-difference approximation for the grid equations at the following boundary-related grid points:

- a) Nodes on the right boundary: $i = M$, $j = 1, 2, \dots, M - 1$.
b) Nodes on the bottom boundary: $j = 0$, $i = 1, 2, \dots, M - 1$.
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- c) Corner grid points: $(M, 0)$ and (M, M) .

Question 3: [Inverse Monotonicity, 6 Points]

Let us define the inverse monotonicity of an operator $\mathcal{L}$ as

$$\mathcal{L}u \leq 0 \Rightarrow u \leq 0.$$

Show that the five-point stencil operator (defined by Eq. (2.7)) is inverse monotone using the above definition.

Question 4: [Norm, 4 Points]

Let

$$V = \left\{ f \in \mathcal{C}^1[0, 1] : f(0) = 0 \right\}.$$

Show that

$$|f|_1 = \left(\int_0^1 |f'(t)|^2 dt \right)^{1/2},$$

defines a norm on V .

Question 5: [Inner Product, 4 Points]

Let $x, y \neq 0 \in X$ where X is an inner product space with inner product $(\cdot, \cdot)$.

- a) Show that if $(x, y) = 0$, i.e., x and y are orthogonal, then $\{x, y\}$ is a linearly independent set.
- b) Extend the results to mutually orthogonal nonzero vectors $\{x_1, x_2, \dots, x_m\}$.

Question 6: [Inner Product, 4 Points]

Let $L^2((0, \infty), e^{-x} dx)$ define the weighted space, i.e.,

$$L^2((0, \infty), e^{-x} dx) = \left\{ u : \int_0^\infty |u(x)|^2 e^{-x} dx < \infty \right\}.$$

Show that

$$a(u, v) = \int_0^\infty e^{-x} u(x) v(x) dx$$

defines an inner product on the weighted space.

Question 7: [Programming, 10 Points]

Consider the Poisson equation

$$\begin{aligned} -\Delta u &= f & \text{in } \Omega, \\ u &= g & \text{on } \Gamma, \end{aligned}$$

where $\Gamma = \partial\Omega$ denotes the boundary of the domain.

We have derived the nine-point stencil for solving the above PDE in the case $\Omega = [0, 1]^2$ with uniform grid spacing.

The right-hand side f and the boundary condition g are chosen such that the exact solution is

$$u(x, y) = x^4 y^5 - 17 \sin(xy).$$

From theory, we expect a convergence rate of order 4, which can be verified numerically. For bonus points, extend the implementation of the five-point stencil to the nine-point stencil and verify the order of convergence.
